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The Trigger Effect of GnRH Agonist Repeated Dose on Intracytoplasmic Sperm Injection Outcomes: A Randomized Single-Blinded Clinical Trial

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ABSTRACT

Purpose: The optimal GnRH agonist (GnRHa) dosing for ovulation triggering remains controversial, particularly regarding oocyte competence versus OHSS mitigation. Here, we analyzed the repeated dose of GnRHa as a trigger of ovulation on intracytoplasmic sperm injection (ICSI) results.

Methods: This single-blinded randomized clinical trial was conducted on 362 ICSI candidates who met the inclusion criteria. The subjects underwent the antagonist protocol and based on the received trigger, were divided into two groups: single dose (group A) and repeated dose (group B) of GnRHa. Demographic and clinical data were collected and analyzed using SPSS-24.

Results: Repeated dosing of GnRH increased M2 oocyte numbers yield by 22.84% ± 6.92% (95% CI: 9.23–25.38, $p = 0.001$). Also, the maturity rate ($p = 0.001$), the number of pronuclear embryos ($p = 0.001$), and the number of good and excellent quality embryos ($p = 0.004$) were higher in group B. Empty follicle syndrome was presented in no cases. Despite the high ovarian response, no OHSS cases were reported (mean oocytes> 17 ± 8). There was no significant difference regarding premature ovulation, use of Cabergoline, chemical and clinical pregnancy, miscarriage, and live birth rates between the 2 study groups.

Conclusion: It seems that the administration of the second dose of GnRHa 12 h after the first dose leads to better oocyte maturation and higher quality embryos.

Trail Registration: IRCT20081007001306N11; https://irct.behdasht.gov.ir/trial/60387

1 | Introduction

Oocyte final maturation is a critical key for fertilization and retrieval results in assisted reproductive technology (ART) [1]. For the first time, in 1990, Gonen et al. [2] introduced gonadotropin-releasing hormone agonist (GnRHa) as an induction of the oocyte final maturation in an antagonist protocol.

Nowadays, a single bolus of GnRHa has become an alternative to human chorionic gonadotrophin (hCG) to reduce its prolonged bioactivity resulting in a sustained luteotropic effect which may induce ovarian hyperstimulation syndrome (OHSS) [3]. Also, ovarian stimulation with final oocyte maturation through GnRHa is more similar to the natural cycle and has a more physiological pattern [4]. The other reasons presented in the studies

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are the poor endometrial preparation and probable lower quality of the embryos during hCG trigger usage [2–4].

On the other hand, as a normal response, with administration of a GnRHa single dose, gonadotropins (FSH and LH) release from the anterior pituitary and have an adverse effect on the corpus luteum [4].

There are some recent studies [3–8] declaring that the dual trigger of GnRHa (repeated dose or together with hCG) may provide more sustained support for the corpus luteum and stronger results. Such benefits have been reported in both normal and poor responders [6–8].

There are fewer studies that have investigated the proper dosage of GnRHa for the trigger effect of ovulation which results in oocyte maturation with a low chance of OHSS [9]. In a natural cycle, the duration of GnRHa trigger is shorter and may not result in intrinsic LH increases to the upper limit or above, which is necessary for the ovulation process, especially in polycystic ovarian syndrome [7–9].

There are several studies on GnRH-a types, doses, and modes of administration on the ICSI results; 0.2 mg triptorelin [10–12] 1 mg Leuprolide acetate [13], 1.5 mg leuprolide-acetate [14], 0.5 mg Buserelin [15–18] and 0.2 mg intranasal Buserelin [19]. Also, Parneix et al. [20] studied the administration of different forms and types of GnRHa (Leuprolide acetate, Triptorelin, Buserelin, and Nafarelin) in different doses and intervals. They concluded that all types of GnRHa trigger sufficiently induced ovulation. In 1994, Shalev et al. [21] reported the successful trigger effect of 0.5 mg triptorelin in ART. Then the next year, they assessed the efficient dosage and concluded that a single 0.1 mg dose of triptorelin was sufficient to induce ovulation in 24 cycles [22] and 1 year later, they published an article proposing that 0.1 mg triptorelin was superior to doses of 0.05 and 0.5 mg for promoting ovulation [23].

Also, in 2016, Voung et al. [24] reported no significant differences in the trigger effect between triptorelin doses of 0.2, 0.3, and 0.4 mg used in oocyte donors with regard to the number of mature oocytes and embryos of higher quality. GnRH agonist triggers oocyte maturation by inducing a surge of both LH and FSH from the anterior pituitary through activation of GnRH receptors. However, this effect is transient. After an initial surge, GnRH receptors on pituitary gonadotropes undergo rapid desensitization and internalization, resulting in a sharp decline in LH secretion within 12–24 h [23, 25]. This decline is particularly problematic for luteal phase support, and in some patients, even the initial LH surge may be insufficient if pituitary responsiveness is impaired. Increasing the dose of GnRHa does not proportionally increase the magnitude or duration of the LH/FSH surge due to this receptor desensitization [25, 26]. This is supported by pharmacodynamic studies showing that higher doses lead to more receptor downregulation, not sustained hormonal stimulation. In contrast, repeated administration of GnRHa (e.g., at 12-hour intervals) may reactivate remaining responsive pituitary cells or temporally extend the window of LH/FSH release, thereby improving final oocyte maturation and luteal steroidogenesis in selected populations, such as suboptimal responders. This strategy is supported by preliminary studies and case reports [1, 18] Based on these variant results, here we assessed triptorelin dual therapy as an ovulation trigger on intracytoplasmic sperm injection (ICSI) results.

2 | Materials and Method

2.1 | Patients and Methods

In this randomized single-blinded clinical trial, all infertile women who met the inclusion criteria and were candidates for intracytoplasmic sperm injection (ICSI) were referred to the first author's private practice and the infertility clinic of Al-Zahra Hospital in Rasht (a city in northern Iran) between January 2022 and January 2023 and were enrolled in the study.

2.2 | Ethical Considerations

The subjects participate in the survey after explaining the study protocol and reassuring them about the confidentiality of all information, no changes in the treatment process, and the imposition of no cost.

The study protocol was approved by the Ethics Committee of the Vice-Chancellor of Research at Guilan University of Medical Sciences (code: IR.GUMS.REC.1400.415, date: 2021-11-24) and registration number: IRCT20081007001306N11. All subjects signed the informed consent forms.

2.3 | Inclusion Criteria

Age ≤ 41 years, Absence of severe male factor (sperm count $< 5 \times 10^6$), no uterine anomalies (fibroids, Müllerian anomaly, adenomyosis, polyps, etc.), no ovarian cysts such as endometrioma, no metabolic diseases (such as diabetes) and any disease with a negative effect on ovulation (such as thyroid disease or hyperprolactinemia), absence of chronic heart, liver and kidney diseases.

2.4 | Exclusion Criteria

Absence of at least 3 dominant follicles on the day of trigger, weak preparation of the endometrium for implantation. The need for extra specific methods to prepare the endometrium, such as increasing Estradiol dosage and platelet-rich plasma (PRP) infusion within the endometrial cavity, or any other interventions.

2.5 | Sample Size and Sampling Methods

The sample size required to investigate the effect of repeated doses of GnRH agonist on ICSI outcomes, with a 95% confidence interval and 80% statistical power, was calculated using the two-sample test based on the study by Deepika et al. [27], resulting in 168 participants per group. Due to the possible loss of the sample, we consider the sample size to be 10% more, and therefore, the number of samples would be 186 subjects in each group (total = 372 people) (Figure 1).

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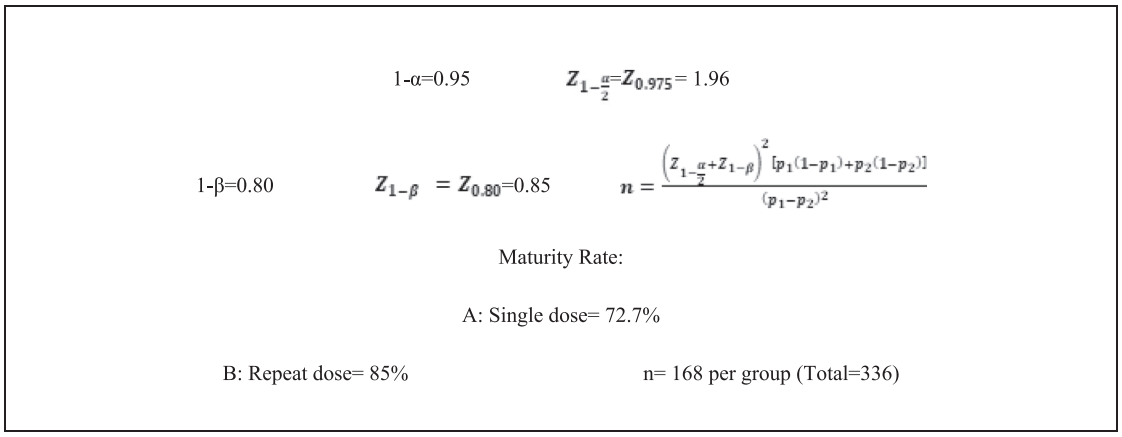


FIGURE 1 | Sample size calculation.

2.6 | Randomization

After informed consent, all subjects were divided into 2 groups: A (GnRH Single dose) and B (GnRH double dose) using the random block method (93 blocks of 4). Randomization was performed using the random allocation software. The results were placed in a separate envelope according to the software list and given to a third person. Then, the envelope was opened for each subject, and the protocol continued according to the desired sequence. The protocol was based on intention to treat (ITT) and single-blinded, as the subjects were not aware of the treatment.

2.7 | Study Protocol

At the beginning, based on the previous cycle pattern, each subject was treated with 10 mg Medroxyprogesterone acetate (MPA) from the 14th day of the menstrual cycle for 10 days or 2 mg Estradiol valerate twice a day, from the 21st day until the first day of the next cycle.

During days 1–3, the subjects underwent transvaginal ultrasound (TVS), and if there was no cyst over 15 mm in diameter in the ovaries, stimulation was started using Gonadotropins. The starting dose of Gonadotropins depended on the cause of infertility, the maternal age, the previous experience and discretion of the attending physician, and at least 2 Pure FSH (Cinnal-F; Cinnagen, Sobhan Daru, Tehran, Iran) was administered for each patient.

GnRH antagonist (Cetronax 250 μg; Ronak, Tehran, Iran) was started in the flexible protocol when the follicles reached at least 13–14 mm in diameter. Gonadotropins and GnRH antagonist continued until the day of the trigger. Follicle growth was followed by TVS. From the second half of the stimulation cycle, HMG ampoule (Menogon,75 IU—Ferring, Switzerland) was added, and medication dosage was adjusted.

On the trigger day, the serum levels of Estradiol, LH, and progesterone were measured. When at least three dominant follicles with a diameter of at least 17 mm were achieved on TVS, the subjects were randomly divided into two groups: the first group (A) with a single injection of 0.2 mg GnRH agonist (Decapeptyl 0.1 mg Ipsen France) 35–38 h before ovum pick up. The second group (B) with a dual GnRH agonist injection of 0.2 mg at 38–35 h before ovum pick up and 0.1 mg 12 h after the first dose. Figure 2 is a schematic view of the induction protocol of ovulation in this survey (Figure 2).

Then, under general anesthesia and with TVS guidance, the follicles were aspirated and assessed at the IVF laboratory, and the metaphase II (MII) oocytes underwent ICSI fertilization, and all obtained embryos were frozen.

During frozen embryo transfer [28] cycle, the receiving uterus was prepared with at least 2 mg Estradiol Valerate 3 times a day from the third day of the menstrual cycle, and the state of the endometrium (pattern and thickness) was assessed by TVS. At least after 2 weeks of Estradiol consumption, if the endometrium thickness was at least 7 mm with a three-line pattern, embryo transfer of day 2 or 3 embryos was performed. Suppository progesterone (Cyclogest 400 mg Actoverco Iran) twice daily was started 2–3 days before embryo transfer and continued up to 10 weeks of gestation. β-HCG was measured about 2 weeks after embryo transfer, and clinical pregnancy was confirmed by detection of the fetal heart on TVS. Also, the patients were followed until miscarriage or live birth.

2.8 | Variables

- Demographics: Age, body mass index, infertility duration, the type of infertility (primary or secondary), the main cause of infertility (male factor, tubal factor, ovarian factor, idiopathic), and anti-Müllerian hormone titer.
- Main variables during study protocol: Dose of gonadotropins, blood level of progesterone, estradiol, and LH on the trigger day
- The primary endpoint: The total number of oocytes, the number of GV, M1, and M2 oocytes, and the maturity rate.
- The second endpoint: The total number of embryos, the number of 2-pronuclear embryos, the number of embryos based on quality excellent and good (A and B), the number of transferred embryos,
- Final results: Empty follicle syndrome, OHSS, Cabergoline administration, maturity rate (the ratio of the number of M2

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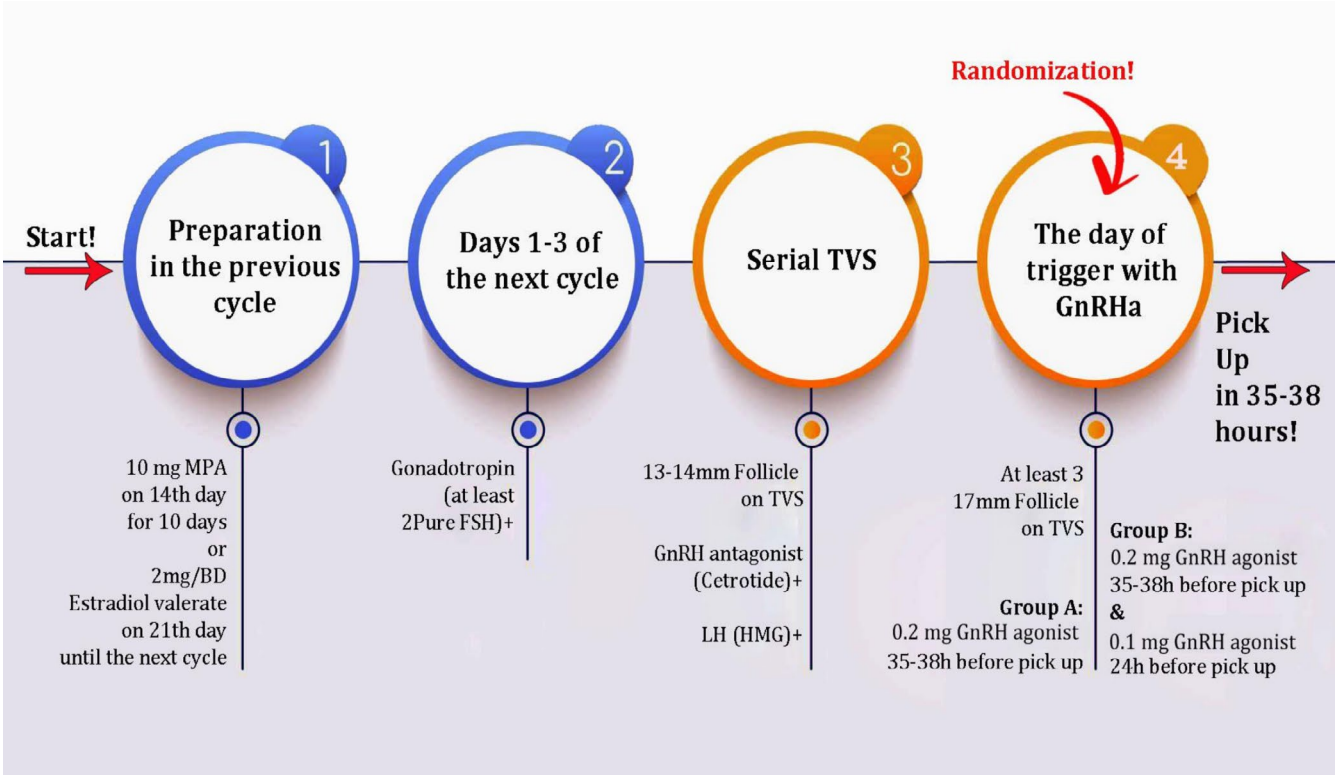


FIGURE 2 | A schematic view of the induction protocol of ovulation in this survey.

oocytes to the total number of obtained oocytes), fertilization rate (the ratio of 2PN embryos to M2 eggs), biochemical pregnancy, clinical pregnancy (fetal heart detection on TVS), miscarriage, and live birth rates.

2.9 | Statistical Analysis

The data were analyzed using SPSS software version 24. In univariate analysis, in order to compare the qualitative variables, the chi-square test and Fisher's exact test were used, and in order to analyze the quantitative variables, the independent *t*-test was used. Also, if the assumptions were not established, the Mann–Whitney *U*-test was used. To determine the effect of repeated doses on outcomes in multivariate analysis, logistic regression was used for qualitative outcomes, and linear regression was used for quantitative outcomes. In order to remove auxiliary or covariant intervening variables from the research results, the statistical method of analysis of covariance was used. The significance level of the tests in this study was considered with $p < 0.05$.

3 | Results

As the Consort flowchart demonstrates, at the first step, 372 participants were included in the study, and then 10 subjects were excluded due to exclusion criteria and lost to follow-up. Finally, 181 patients enrolled in both the A and B study groups (Figure 3).

The baseline characteristics of all patients are presented in Table 1. The mean age of subjects was 34 ± 6.5 years, and the mean titer of Anti-Mullerian hormone was 4.45 ± 3.1 ng/mL. The most common cause of infertility was ovarian factor (58.28%), and most subjects suffered from primary infertility (74.31%). There was no statistically significant difference between the two study groups based on the demographic characteristics (Table 1).

Duration of the controlled ovarian stimulation and the mean total dose of gonadotropins among all subjects showed no significant difference between groups A and B. Also, progesterone, estradiol, and LH levels on the trigger day presented the same pattern of distribution. For the prevention of ovarian hyperstimulation syndrome (OHSS) in patients with high levels of serum estradiol (> 3000 pg/mg), cabergoline was administered in 25 subjects in group A and 15 cases in group B (14% and 8.8% respectively; $p > 0.005$). In this study, no OHSS showed up (Table 2).

During the induction of ovulation, 1 case in group A showed premature ovulation, and 6 subjects in group B were excluded due to missing data. So, 180 cases in group A and 175 cases in group B underwent oocyte retrieval. Also, 1 patient in group A and 4 patients in group B were excluded due to no ovum, and finally, the embryos were transferred in 179 subjects in group A and 171 cases in group B.

Table 3 demonstrates the oocyte retrieval results in both study groups. No empty follicle was reported. As the data shows, the number of M2 oocytes was significantly more in group B (13 ± 6 vs. 10 ± 5; $p = 0.001$). Also, the maturation rate was 66.02 ± 18.27 and 73.35 ± 17.69 in groups A and B respectively ($p = 0.001$). In the same way, the number of fertilized oocytes ($p = 0.002$), two pronuclear oocytes ($p = 0.001$), and embryos with good and

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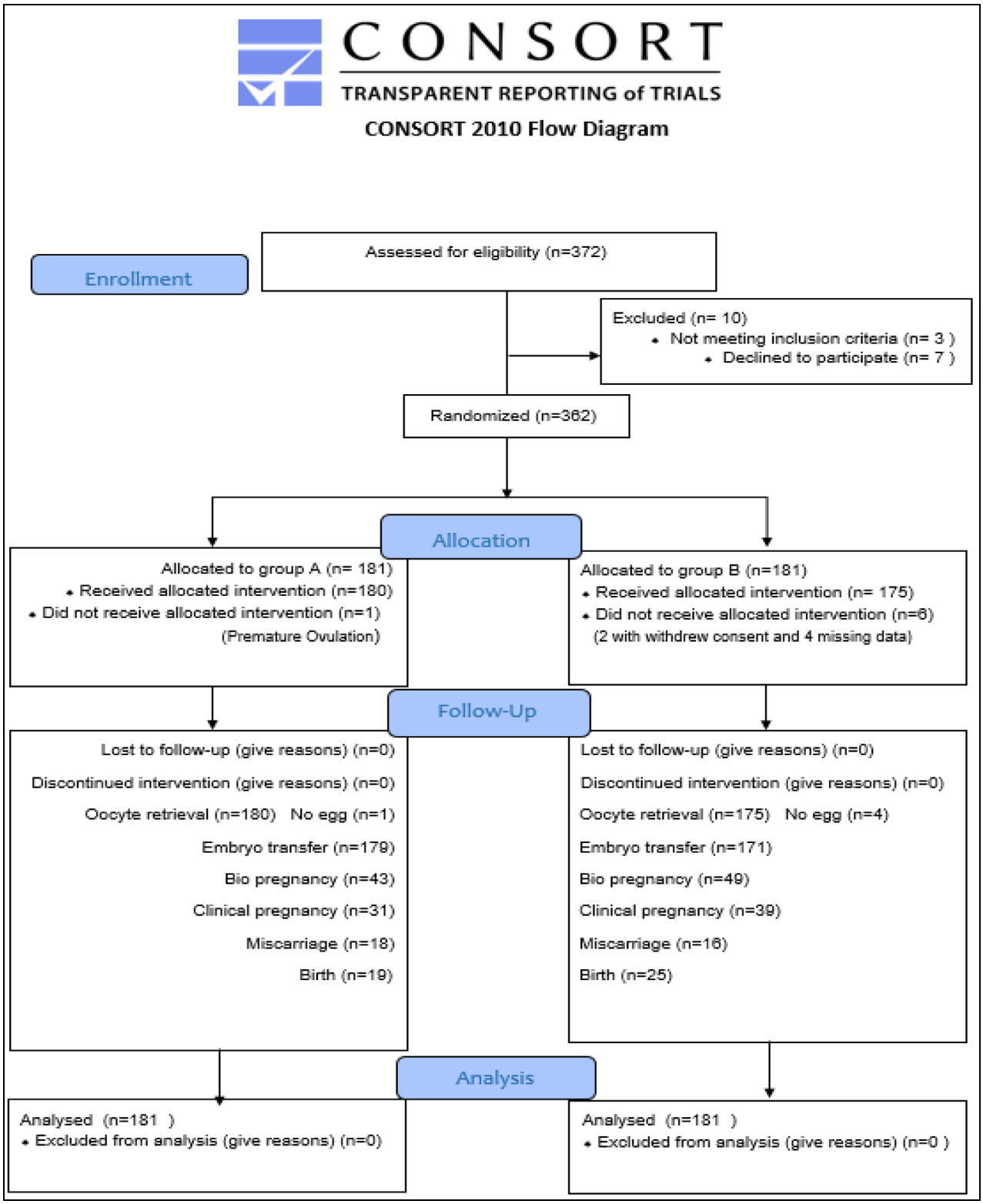


FIGURE 3 | The study flow chart.

excellent quality ($p = 0.004$) was higher in group B (Table 3). Repeated dosing of GnRH increased M2 oocyte numbers yield by 22.84% ± 6.92% (95% CI: 9.23–25.38, $p = 0.001$). On the other hand, as Table 4 shows, the results of embryo transfer did not statistically significantly differ in terms of pregnancy, miscarriage, or live birth rates (Table 4).

4 | Discussion

4.1 | Principles in the Context

Mimicking the effect of LH surge, hCG is traditionally used as a trigger of oocyte maturation during IVF. But nowadays,

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TABLE 1 | Patient's demographic features in both study groups.

	Group A	Group B	
	***n* (181)**	***n* (181)**	***p***
Maternal age (years)	34 ± 6	34 ± 7	0.767
Body mass index (kg/m^2)	26.6 ± 4.28	26.47 ± 4.01	0.707
Infertility duration (years)	5 ± 3	5 ± 4	0.483
Primary infertility (*n*%)	142 (79.3%)	127 (74.3%)	0.262
Secondary infertility (*n*%)	37 (20.7%)	44 (25.7%)	0.428
Male factor (*n*%)	28 (15.6%)	33 (19.3%)	
Tubal factor (*n*%)	25 (14%)	15 (8.8%)	
Ovarian factor (*n*%)	107 (59.8%)	104 (60.8%)	
Idiopathic infertility (*n*%)	19 (10.6%)	19 (11.1%)	
Anti Mulerian hormone titer (ng/mL)	4.3 ± 2.4	4.6 ± 3.5	0.278

Note: $p < 0.05$.

TABLE 2 | Comparison of the cycle and hormonal characteristics in both study groups.

	Groups		
	A (*n* = 181)	**B (*n* = 181)**	**p**
Gonadotropin total dosage (IU)	3343 ± 905	3435 ± 819	0.318
Progesterone level on trigger day (ng/mL)	2.22 ± 0.61	1.68 ± 0.68	0.362
Estradiol level on trigger day (pg/mL)	3744 ± 2447	3727 ± 2398	0.949
LH level on trigger day (ng/mL)	1.31 ± 1.09	1.30 ± 1.06	0.910
Duration of gonadotropin adminstration (days)	10 ± 2	10 ± 1	0.671
Cabergoline administration (*n*%)	25 (14%)	15 (8.8%)	0.127
OHSS rate (*n*%)	0 (0%)	0 (0%)	—

Note: $p < 0.05$.

TABLE 3 | Comparison of the oocyte retrieval results among the two study groups.

	Groups		
	A (*n* = 180)	**B (*n* = 175)**	***p***
Number of retrieval oocytes	16 ± 7	18 ± 9	0.028
Premature ovulation rate	1 (0.55%)	0 (0%)	0.328
Empty follicle syndrome rate	0 (%)	0(%)	—
Number of type GV oocytes	2 ± 2	2 ± 2	0.526
Number of type M1 oocytes	2 ± 2	2 ± 2	0.344
Number of type M2 oocytes	10 ± 5	13 ± 6	0.001*
Fertilization rate	67.89 ± 22.29	71.33 ± 24.04	0.121
Number of 2 pronuclear(2 PN) embryo	6.77 ± 3.85	8.91 ± 5.14	0.001*
Oocyte maturation rate	66.02 ± 18.27	73.35 ± 17.69	0.001*
Number of fertilized oocytes	7.36 ± 4.19	9.14 ± 5.21	0.002*
Embryos with good & excellent quality	5.98 ± 3.4	7.39 ± 4.39	0.004*

*$p < 0.05$.

TABLE 4 | Comparison of the embryo transfer results among the two study groups.

	Groups		
	A (*n* = 180)	**B (*n* = 175)**	***p***
Number of transferred embryos	2.37 ± 0.56	2.45 ± 0.61	0.125
Biochemical pregnancy rate	43 (23.88%)	49 (28%)	0.326
Clinical pregnancy rate	31 (17.22%)	39 (22.28%)	0.199
Miscarriage rate	18 (10%)	16 (9.14%)	0.968
Live birth rate	19 (10.6%)	25 (14.28%)	0.333

Note: $p < 0.05$.

due to the widespread use of the antagonist protocol, GnRHa has been used as a substitute with a lower risk of OHSS and good results. Here we studied the trigger effect of GnRHa on ICSI results and assessed whether using repeated dosage may result in higher quality of oocyte and better pregnancy outcomes.

4.2 | Clinical and Research Implications

The maternal age and BMI, the main cause of infertility, the AMH level, and the other demographic features were not statistically different between the two groups, revealing that the subjects were homogeneous. Similar to Weidenbaum et al. [29]

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and Blakemore et al. [30], our results showed the total number of obtained oocytes in the GnRH agonist repeated dose group (B) was significantly higher than in the single dose group (A). Also, similar to Deepika et al. [27], the number of M2 eggs in group B was significantly higher ($p < 0.001$). Our other findings revealed a significant difference between the two study groups regarding the number of 2 PN oocytes ($p < 0.001$) and the number of good and excellent quality embryos ($p = 0.004$). On the other hand, in the same way as some studies such as Aflatonian 12et al [31] and Blakemore et al. [30], our results showed no statistically significant difference between the two study groups regarding the number of M1 and GV eggs ($p = 0.344$ and $p = 0.526$, respectively). Also, Vuong et al. [30] concluded no significant differences between triptorelin doses of 0.2, 0.3, and 0.4 mg used for ovulation trigger in oocyte donors regarding the mature oocytes and high-quality embryos.

The probable theory for the better results of GnRHa repeated dose is based on its actual pathophysiology. As we know, unlike hCG trigger, GnRHa results in both LH and FSH surges and is more similar to the natural cycle, but with a fast LH surge with a short duration [7, 8]. This short LH surge duration may result in poor oocyte maturation and insufficient support for the corpus luteum [8, 9]. So, it has been proposed that GnRHa repeated dose may play an adjuvant role, resulting in longer duration of LH and FSH serum levels and better trigger effect on oocyte maturation in terms of number and quality [6].

Traditionally, HCG has been used for final oocyte maturation with a higher pregnancy rate [32]. Also, due to its long half-life, the most frequent adverse effects were higher steroid levels during the luteal phase and a decrease in endometrial receptivity, and lower embryo quality [2, 3]. On the other hand, in recent decades, GnRHa has been proposed for oocyte final maturation in GnRH antagonist cycles [3, 4]. It results in shorter duration and smaller amplitude of FSH and LH than the levels resulting from HCG administration. Also, the serum levels of FSH and LH are similar to the natural mid-cycle LH surge, resulting in a lower risk of OHSS. Some studies, such as Benadiva et al. [33] and Sukur et al. [34] have reported the excellent effect of GnRHa trigger in hyper responders, but there is no evidence on normal responders and double doses of GnRHa trigger. Here, we could not determine the LH levels during Triptorelin trigger administration, but we assessed the final results on oocyte qualities and embryos. Based on Lu et al. [35] the dual-trigger protocol capitalizes on the GnRH-agonist effect on FSH-LH which leads to a physiological mimic of natural ovulation that results in LH receptor upregulation, maturation of nuclei, and cumulus expansion. The LH surge precipitates GnRH-receptor internalization and desensitization of gonadotropins, which results in a higher drop in endogenous LH during the early luteal phase and an insufficient luteal phase. So, with a low-dose hCG, corpus luteum function sustains, and oocyte maturation improves.

In our study, as in Deepika et al. [27], maturation rate showed a statistically significant difference between the two study groups (higher in group B; $p < 0.001$), but the rate of fertilization and implantation showed no significant difference between the two groups ($p = 0.121$ and $p = 0.749$, respectively).

One of the possible theories of GnRHa trigger effect is based on the growth factors' critical role during maturation. Amphiregulin and other epidermal growth factors rapidly increase in follicular fluid in response to LH/hCG and are thought to play a significant role in oocyte maturation mediating intra-follicular LH effects. Amphiregulin levels in follicular fluid after GnRH agonist stimulation are much lower than in hCG-stimulated follicles and similar to the levels in a natural cycle [27, 30, 31]. Also, it has been reported that the restart of meiosis is at 18 h after the LH surge, and so, to achieve oocyte maturation with higher quality, the LH serum level should be maintained for 14–27 h. Some studies, such as Deepika et al. [27] and Aflatoonian et al. [31] claimed that a repeat dose of GnRHa could maintain the LH surge. So we added the second dose of GnRH 12 h after the first dose. Also, the discrepancy between the increased quality of embryos and the fact that there is no significant difference in the pregnancy rate may be explained by the small sample size and further clinical trials may be needed for more exact results.

Vascular endothelial growth factor (VEGF) is also significantly reduced in follicular fluid after GnRH agonist stimulation and VEGF mRNA expression in granulosa cells is reduced compared to hCG stimulation [36]. VEGF is one of the important vasoactive substances and affects, to some extent by modulating the permeability of endothelial cells. Significant reduction in VEGF and vascular permeability after GnRHa administration plays a major role in preventing OHSS. In fact, it has been proposed that GnRH agonist creates a more anti-angiogenic environment that may lead to impaired corpus luteum function and prevent OHSS [31, 36].

Interestingly, in the same way, no OHSS was reported in our study. Aflatonian et al. [31] and Weidenbaum et al. [29] reported the rate of OHSS to be similar in both single-dose and repeated-dose groups, but Blakemore et al. [30] declare that the rate of moderate and severe OHSS was significantly lower in the Triptorelin repeated dose group. The other hypothesis is that the high carbohydrate content of hCG results in an LH-like activity with a long half-life that leads to continuous LH receptor stimulation. On the other hand, GnRHa increases LH and FSH release with a shorter half-life, resulting in a significant reduction in the risk of OHSS.

In fact, follicular fluid after GnRHa stimulation shows significantly higher levels of LH and FSH than hCG trigger, while progesterone levels decrease, attributed to the lack of LH stimulation in luteal cells. However, estrogen, Inhibin-A, and Inhibin-B levels are similar in both triggers [31, 36]. These differences in follicular fluid dynamics may indicate a greater difference between the signal required for oocyte maturation versus the signal required for ovulation. Although maturation and ovulation are usually two closely related events, they may require slightly different signals [31, 36, 37].

Empty follicle syndrome was not seen in any of the study groups of this survey. This syndrome is one of the IVF process consequences that result in cycle cancellation. The exact pathophysiology is still unclear. In some studies, it has been reported as a complication of using a single dose of GnRHa as the trigger [27, 30].

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The pituitary gland is the target of GnRHa, and disruption of the hypothalamus–pituitary–ovary axis may result in incomplete final oocyte maturation and empty follicle syndrome [36, 37]. One hypothesis is that administration of the second dose of GnRHa 12 h after the first dose may improve the duration and amplitude of the LH surge and result in oocyte maturation. However, in our participants, no empty follicle syndrome was reported in either group to support or reject this hypothesis.

Our findings showed that the rate of chemical and clinical pregnancies and live birth rate were similar in both study groups. Deepika et al. [27] reported that although the chance of clinical pregnancy per patient in the repeated dose group was higher than in the single dose group (OR = 0.56, 0.56), it was not statistically significant, which may be due to the low sample size. Also, in our study, although the number of abortions in group A was clinically higher than in the repeated dose group, this difference was not statistically significant ($p = 0.666$). As a strong guideline, the ESHRE guideline declares that GnRH agonist may be an effective trigger in ICSI candidate women at risk of OHSS especially poor responders, and also, repeated GnRH agonist injections, alone or in addition to progesterone, could be effective for luteal phase support, but, there is no clue of the GnRH repeated dose effect on ICSI final results in normal responders and more clinical trials may be necessary to investigate such a hypothesis.

Although retrospective LH measurements were not available in our study, we have expanded the theoretical framework underpinning dual-trigger protocols by incorporating relevant endocrinologic and molecular insights; 1-Dynamics of GnRH Agonist Triggering: A GnRH agonist, through GnRH receptor activation, induces a short-lived and rapid surge of both LH and FSH. The pituitary GnRH receptor reserve and GnRH receptor downregulation occur due to internalization and desensitization mechanisms, which regulate the magnitude and duration of this surge that [29, 30]. The resulting biphasic surge, with a sharp LH peak and a smaller but biologically relevant FSH peak, closely mimics the endogenous mid-cycle surge [28, 34]. FSH has been shown to play a critical role in oocyte nuclear maturation, cumulus expansion, and granulosa cells' LH receptor expression [29, 30]. 2-Gonadotrope Desensitization and the Need for hCG Co-Administration: Continuous stimulation with GnRH agonist results in rapid internalization of GnRH receptors, leading to gonadotrope desensitization and suppression of endogenous LH support in the luteal phase [8, 9]. This is why a dual-trigger protocol includes a small dose of hCG, which provides longer-acting LH-like activity, supporting the resumption of meiosis, luteinization, and early luteal support [9, 28]. hCG acts directly on LH receptors in granulosa-lutein cells and promotes progesterone synthesis. 3- Synergistic Action and Clinical Rationale: The dual-trigger aims to capitalize on the FSH surge for oocyte maturation and cumulus expansion, which hCG alone does not provide, while ensuring adequate LH-like support through low-dose hCG to sustain corpus luteum function and implantation potential [10, 11, 34].

4.3 | Strengths and Limitations

The strength of our survey was the minimized bias because of the study protocol, a single-blinded controlled trial. On the study limitations, unfortunately, we did not measure the LH level after administration of the first and second doses of agonist trigger prior to oocyte retrieval due to the additional cost. While our study did not include serum LH measurements after triggering, several clinical parameters indirectly point to adequate and sustained LH activity in the dual trigger group:

- A higher proportion of mature oocytes (MII) indicates effective LH action necessary for final oocyte maturation.
- The better embryo morphology observed likely reflects improved oocyte quality, which depends in part on optimal LH-driven cytoplasmic and nuclear maturation.
- Additionally, higher fertilization rates and oocyte yield relative to follicle count may signal more competent oocytes, indirectly supporting the adequacy of the LH surge.

5 | Conclusion

It seems that the administration of the second dose of GnRHa 12 h after the first dose leads to better oocyte maturation and higher-quality embryos.

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Ethics Statement

The study protocol was approved by the Ethics Committee of the Guilan University of Medical Sciences (code: IR.GUMS.REC.1400.415, date: 2021-11-24) and registration number: IRCT20081007001306N11.

Consent

The Study participants, after explaining the study protocol and being reassured about the confidentiality of all information and no changes in the treatment process and imposition of no cost, signed the informed consent forms.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data available on request due to privacy/ethical restrictions. The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions. Although the data are anonymized, the Ethics Committee has not given permission for the data to be shared because they contain information that could be used to identify individuals when matched with other information. If the data need to be provided to a third party, the authors will again need to obtain permission from the Ethics Committee.

References

1. K. Chen, C. Zhang, L. Chen, Y. Zhao, and H. Li, "Reproductive Outcomes of Dual Trigger Therapy With GnRH Agonist and hCG Versus hCG Trigger in Women With Diminished Ovarian Reserve: A Retrospective Study," *Reproductive Biology and Endocrinology* 22, no. 1 (2024): 35.

14470578, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rmb2.12690, Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2. Y. Gonen, H. Balakier, W. Powell, and R. F. Casper, "Use of Gonadotropin-Releasing Hormone Agonist to Trigger Follicular Maturation for In Vitro Fertilization," *Journal of Clinical Endocrinology & Metabolism* 71, no. 4 (1990): 918–922.

3. C. Zhou, X. Yang, Y. Wang, et al., "Ovulation Triggering With hCG Alone, GnRH Agonist Alone or in Combination? A Randomized Controlled Trial in Advanced-Age Women Undergoing IVF/ICSI Cycles," *Human Reproduction* 37, no. 8 (2022): 1795–1805.

4. B. Tu, H. Zhang, L. Chen, et al., "Co-Administration of GnRH-Agonist and hCG (Double Trigger) for Final Oocyte Maturation Increases the Number of Top-Quality Embryos in Patients Undergoing IVF/ICSI Cycles," *Journal of Ovarian Research* 17, no. 1 (2024): 137.

5. F. Gao, Y. Wang, M. Fu, et al., "Effect of a "Dual Trigger" Using a GnRH Agonist and hCG on the Cumulative Live-Birth Rate for Normal Responders in GnRH-Antagonist Cycles," *Frontiers in Medicine* 8 (2021): 683210.

6. S. H. Hashemi, M. Hafezi, A. Arabipoor, M. Zareei, S. Vesali, and P. Eftekhari-Yazdi, "Can We Harvest More Mature Oocytes by Repeating Gonadotropin-Releasing Hormone Agonist Doses in Polycystic Ovarian Syndrome Patients at Risk of OHSS in Antagonist Cycles? A Randomised Clinical Trial," *International Journal of Fertility and Sterility* 18, no. Suppl 1 (2024): 48–54.

7. I. F. Kozlowski, V. B. da Rosa, C. D. de Souza Francisquini, and A. Schuffner, "Dual Trigger and the Impact on Oocyte Quality and Embryo Development: A Brazilian Cohort," *JBRA Assisted Reproduction* 27, no. 4 (2023): 629.

8. L. Dong, F. Lian, H. Wu, et al., "Reproductive Outcomes of Dual Trigger With Combination GnRH Agonist and hCG Versus Trigger With hCG Alone in Women Undergoing IVF/ICSI Cycles: A Retrospective Cohort Study With Propensity Score Matching," *BMC Pregnancy and Childbirth* 22, no. 1 (2022): 583.

9. B. Shakerian, E. Turkgeldi, S. G. Cekic, S. Yildiz, I. Keles, and B. Ata, "Dual Trigger Compared With Human Chorionic Gonadotropin Alone and Effects on Clinical Outcome of Intracytoplasmic Sperm Injection," *International Journal of Fertility & Sterility* 15, no. 4 (2021): 294–299.

10. D. Bodri, J. J. Guillen, A. Galindo, D. Mataro, A. Pujol, and O. Coll, "Triggering With Human Chorionic Gonadotropin or a Gonadotropin-Releasing Hormone Agonist in Gonadotropin-Releasing Hormone Antagonist-Treated Oocyte Donor Cycles: Findings of a Large Retrospective Cohort Study," *Fertility and Sterility* 91 (2009): 365–371.

11. E. R. Hernandez, J. L. Gomez-Palomares, and E. Ricciarelli, "No Room for Cancellation, Coasting, or Ovarian Hyperstimulation Syndrome in Oocyte Donation Cycles," *Fertility and Sterility* 91 (2009): 1358–1361.

12. E. G. Papanikolaou, W. Verpoest, H. Fatemi, B. Tarlatzis, P. Devroey, and H. Tournaye, "A Novel Method of Luteal Supplementation With Recombinant Luteinizing Hormone When a Gonadotropin-Releasing Hormone Agonist Is Used Instead of Human Chorionic Gonadotropin for Ovulation Triggering: A Randomized Prospective Proof of Concept Study," *Fertility and Sterility* 95 (2011): 1174–1177.

13. L. Engmann, A. DiLuigi, D. Schmidt, J. Nulsen, D. Maier, and C. Benadiva, "The Use of Gonadotropin-Releasing Hormone (GnRH) Agonist to Induce Oocyte Maturation After Cotreatment With GnRH Antagonist in High-Risk Patients Undergoing In Vitro Fertilization Prevents the Risk of Ovarian Hyperstimulation Syndrome: A Prospective Randomized Controlled Study," *Fertility and Sterility* 89, no. 1 (2008): 84–91.

14. J. C. Castillo, M. Dolz, E. Bienvenido, L. Abad, E. M. Casañ, and F. Bonilla-Musoles, "Cycles Triggered With GnRH Agonist: Exploring Low-Dose HCG for Luteal Support," *Reproductive Biomedicine Online* 20, no. 2 (2010): 175–181.

15. P. Humaidan, H. E. Bredkjaer, L. Bungum, et al., "GnRH Agonist (Buserelin) or hCG for Ovulation Induction in GnRH Antagonist IVF/ICSI Cycles: A Prospective Randomized Study," *Human Reproduction* 20 (2005): 1213–1220.

16. P. Humaidan, H. Ejdrup Bredkjaer, L. G. Westergaard, and C. Yding Andersen, "1500 IU Human Chorionic Gonadotropin Administered at Oocyte Retrieval Rescues the Luteal Phase When Gonadotropin-Releasing Hormone Agonist Is Used for Ovulation Induction: A Prospective, Randomized, Controlled Study," *Fertility and Sterility* 93 (2010): 847–854.

17. P. Humaidan, N. P. Polyzos, B. Alsbjerg, et al., "GnRHa Trigger and Individualized Luteal Phase hCG Support According to Ovarian Response to Stimulation: Two Prospective Randomized Controlled Multi-Centre Studies in IVF Patients," *Human Reproduction* 28 (2013): 2511–2521.

18. P. Humaidan, L. Bungum, M. Bungum, and C. Yding Andersen, "Rescue of Corpus Luteum Function With Peri-Ovulatory HCG Supplementation in IVF/ICSI GnRH Antagonist Cycles in Which Ovulation Was Triggered With a GnRH Agonist: A Pilot Study," *Reproductive Biomedicine Online* 13 (2006): 173–178.

19. C. Pirard, E. Loumaye, P. Laurent, and C. Wyns, "Contribution to More Patient- Friendly ART Treatment: Efficacy of Continuous Low-Dose GnRH Agonist as the Only Luteal Support–Results of a Prospective, Randomized, Comparative Study," *International Journal of Endocrinology* 2015 (2015): 727569.

20. I. Parneix, J. C. Emperaire, A. Ruffie, and P. Parneix, "Comparison of Different Protocols of Ovulation Induction, by GnRH Agonists and Chorionic Gonadotropin," *Gynécologie, Obstétrique & Fertilité* 29 (2001): 100–105.

21. E. Shalev, Y. Geslevich, and M. Ben-Ami, "Induction of Pre-Ovulatory Luteinizing Hormone Surge by Gonadotrophin-Releasing Hormone Agonist for Women at Risk for Developing the Ovarian Hyperstimulation Syndrome," *Human Reproduction* 9 (1994): 417–419.

22. E. Shalev, Y. Geslevich, M. Matilsky, and M. Ben-Ami, "Induction of Pre-Ovulatory Gonadotrophin Surge With Gonadotrophin-Releasing Hormone Agonist Compared to Pre-Ovulatory Injection of Human Chorionic Gonadotrophins for Ovulation Induction in Intrauterine Insemination Treatment Cycles," *Human Reproduction* 10 (1995): 2244–2247.

23. E. Shalev, M. Matilsky, and Y. Geslevich, "Dose Response Study Using Tryptoreline to Induce Ovulation After Follicular Stimulation With HMG," *Gynecological Endocrinology* 10 (1996): 32.

24. T. N. Vuong, M. T. Ho, T. D. Ha, H. T. Phung, G. B. Huynh, and P. Humaidan, "Gonadotropin-Releasing Hormone Agonist Trigger in Oocyte Donors Co-Treated With a Gonadotropin-Releasing Hormone Antagonist: A Dose-Finding Study," *Fertility and Sterility* 105, no. 2 (2016): 356–363.

25. P. M. Conn and W. F. Crowley, Jr., "Gonadotropin-Releasing Hormone and Its Analogues," *New England Journal of Medicine* 324, no. 2 (1991): 93–103.

26. G. Griesinger, K. Diedrich, P. Devroey, and E. M. Kolibianakis, "GnRH Agonist for Triggering Final Oocyte Maturation in the GnRH Antagonist Ovarian Hyperstimulation Protocol: A Systematic Review and Meta-Analysis," *Human Reproduction Update* 12, no. 2 (2006): 159–168.

27. K. Deepika, P. Baiju, P. Gautham, R. Suvarna, V. Arveen, and R. Kamini, "Repeat Dose of Gonadotropin-Releasing Hormone Agonist Trigger in Polycystic Ovarian Syndrome Undergoing In Vitro Fertilization Cycles Provides a Better Cycle Outcome-A Proof-of-Concept Study," *Journal of Human Reproductive Sciences* 10, no. 4 (2017): 271–280.

28. N. Chabbert-Buffeta, D. C. Skinner, A. Caraty, and P. Bouchard, "Neuroendocrine Effects of Progesterone," *Steroids* 65, no. 10–11 (2000): 613–620.

29. E. M. Weidenbaum, J. K. Blakemore, and J. A. Grifo, "Analysis of the Effect of a Delayed Second Dose of Gonadotropin Releasing Hormone-Agonist (GnRH-a) on Oocyte and Blastocyst Quality and Risk of

14470578, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rmb2.12690, Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ovarian Hyperstimulation Syndrome (OHSS)," *Fertility and Sterility* 112, no. 3 (2019): e215.

30. J. Blakemore, D. McCulloh, and J. Grifo, "Double Duty?: Impact of a Delayed Dose of Gonadotropin-Releasing Hormone Agonist (GnRH-A) After a Dual Trigger on Oocytes Retrieved and Risk of Ovarian Hyperstimulation Syndrome (OHSS)," *Fertility and Sterility* 110, no. 4 (2018): e313–e314.

31. A. Aflatoonian, F. Haghighi, M. Hoseini, and S. Haghdani, "Does the Repeat Dose of Gonadotropin-Releasing Hormone Agonist Trigger in Polycystic Ovarian Syndrome Improve In Vitro Fertilization Cycles Outcome? A Clinical Trial Study," *International Journal of Reproductive BioMedicine* 18, no. 7 (2020): 485–490.

32. M. A. Youssef, F. Van der Veen, H. G. Al-Inany, et al., "Gonadotropin-Releasing Hormone Agonist Versus HCG for Oocyte Triggering in Antagonist-Assisted Reproductive Technology," *Cochrane Database of Systematic Reviews* 2014, no. 10 (2014): CD008046.

33. C. Benadiva and L. Engmann, "Luteal Phase Support After Gonadotropin-Releasing Hormone Agonist Triggering: Does It Still Matter?," *Fertility and Sterility* 109 (2018): 763–767.

34. Y. E. Sukur, B. Ozmen, E. D. Ozdemir, et al., "Final Oocyte Maturation With Two Different GnRH Agonists in Antagonist Co-Treated Cycles at Risk of Ovarian Hyperstimulation Syndrome," *Reproductive Biomedicine Online* 34 (2017): 5–10.

35. X. Lu, Q. Hong, L. Sun, et al., "Dual Trigger for Final Oocyte Maturation Improves the Oocyte Retrieval Rate of Suboptimal Responders to Gonadotropin-Releasing Hormone Agonist," *Fertility and Sterility* 106, no. 6 (2016): 1356–1362.

36. B. P. Jones, A. Al-Chami, X. Gonzalez, et al., "Is Oocyte Maturity Influenced by Ovulation Trigger Type in Oocyte Donation Cycles?," *Human Fertility* 24, no. 5 (2021): 360–366.

37. Y. Wang, Y.-C. Yi, H.-F. Guu, et al., "GnRH Agonist-Only Trigger, Compared to Dual Trigger, Reduces Oocyte Retrieval Rate in High Responders Without Affecting Cumulative Live Birth Rate," *Frontiers in Endocrinology* 15 (2024): 1461317.

14470578, 2025, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rmb2.12690, Wiley Online Library on [28/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License